Lichen Zhu

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Education

Duke University — Durham, NC
M.S. in Electrical and Computer Engineering — *Sep. 2025 – Jul. 2027*

Xi'an Jiaotong-Liverpool University / University of Liverpool — Suzhou, China
B.Sc. in Financial Mathematics, First Class Honours — *Sep. 2021 – Jul. 2025*

Research Experience

Duke University, Department of Electrical and Computer Engineering — Durham, NC
Research Assistant, Center for Computational Evolutionary Intelligence (CEI) — *Sep. 2025 – Present*

- Conduct research on multimodal large language models, long-context memory, video understanding, and efficient AI systems with Yueqian Lin, Yudong Liu, and Prof. Yiran Chen.
- Develop methods and evaluation frameworks for efficient vision-language model inference, long-term interaction memory, and robust multimodal reasoning.

Hefei University of Technology — Hefei, China
Research Assistant — *Jun. 2026 – Present*

- Conduct research on long-form video understanding and efficient video reasoning with Prof. Yanbin Hao and Dr. Shuo Wang.

Publications

Query-Conditioned Evidential Keyframe Sampling for MLLM-Based Long-Form Video Understanding
*Y. Wang, *__L. Zhu__, Y. Lin, Y. Liu, J. Zhang, H. Li, Y. Chen
*Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)**, 2027*

Will My Assistant Remember My Allergy? What Personal LLM Assistants Forget When Conversation Memory Is Compressed
__L. Zhu__, Y. Lin, Y. Wang, Y. Liu, H. Li, Y. Chen
***HumanSys 2026**, Oral Presentation (co-located with ACM MobiCom 2026) arXiv:2609.05767.*

When No Answer Is Correct: Diagnosing Absent Answer Detection for MLLMs in Video Understanding
Y. Wang, Y. Lin, __L. Zhu__, Y. Liu, H. Li, Y. Chen
*The 2026 Conference on Empirical Methods in Natural Language Processing **(EMNLP)**. arXiv:2606.08239.*

Selected Manuscripts

Semantic-Unit Aggregation for Streaming Interaction Memory
__L. Zhu__, Y. Lin, Y. Wang, H. Li, Y. Chen
Under review at AAAI 2027.

Stable Scores, Unstable Answers: Frame Phase and Option Order in Video Multiple-Choice Evaluation
__L. Zhu__, Y. Lin, Y. Wang, H. Li, Y. Chen
Under review at ICASSP 2027.

Knowing When to Trust a Prior: Reliability-Gated Cue Fusion for Video Gaze Prediction
__L. Zhu__, Y. Wang, Y. Lin, H. Li, Y. Chen
Under review at ICASSP 2027.

Coupled but late: Turn-taking between full-duplex speech models in unscripted dialogue
__L. Zhu__, Y. Lin, Y. Wang, H. Li, Y. Chen
Under review at ICASSP 2027.

StyleVAR: Controllable Image Style Transfer via Visual Autoregressive Modeling
L. Jing, D. Zhang, P. Li, __L. Zhu__, Y. Xu, H. Xing
Preprint arXiv:2604.21052.

Additional Experience

Duke University — Durham, NC
Teaching Assistant – Modern Analytics & Machine Intelligence — *Oct. 2026 – Dec. 2026*

- Support DECISION 546Q: Modern Analytics: Data, Prediction, and Actions and DECISION 894: Transforming Tech Analytics with Machine Intelligence.
- Provide office hour support and grade assignments covering machine learning, deep learning, and generative models implemented with Python and PyTorch.

Duke University — Durham, NC
Teaching Assistant – DECISION 520Q: Data Science for Business — *Sep. 2026 – Oct. 2026*

– Supported an MQM data science course through weekly office hours, grading, and technical guidance on R and Python programming, statistical analysis, case studies, and final projects.

China Construction Bank Hefei, China

Financial Technology Intern *Jul. 2022 – Sep. 2022*

– Engineered behavioral features from large-scale transaction data using SQL and Python for downstream financial analytics.

Honors & Awards

Duke University Master's Scholarship, $18,000.

XJTLU Outstanding Graduate, 2025.

National Inspirational Scholarship, Ministry of Education of China, 2024.

XJTLU Outstanding Student, 2023–2024.

Academic Service

Reviewer: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2027.

Technical Skills

Programming: Python (PyTorch, NumPy, Pandas), SQL, R, MATLAB, Java
Research Areas: Multimodal Large Language Models, Vision-Language Models, Long-Form Video Understanding, Long-Term Memory for LLM Agents, Efficient AI Systems
Mathematical Foundations: Probability and Statistics, Optimization, Stochastic Processes, Monte Carlo Methods
Languages: Mandarin Chinese (Native), English (Fluent)